

to support modem and data card applications. System-level power is controlled and managed through NXP chip. Triquint provides three power

amplifiers, each working on different frequencies and providing highly integrated linear amplifier, duplexer and coupler. For communications func-

tion, Skyworks SKY77340 provides power amplification in quad-band GSM/EDGE.

Battery

iPhone 3G faces the same battery limitations as other smart phones when used for the Internet on a 3G network. It uses a lithium-ion polymer battery rated at 3.7V and 1150 mAh (milliampere-hours). The battery is not soldered, unlike the battery in 2G version, making it user-replaceable. The battery of iPhone 3GS is rated at 3.7V and 1219 mAh, which is just 6 per cent higher than the battery of iPhone 3G. Linear Technology's LTC4088-2 takes care of battery charging and USB power control.

Other ICs

Infineon provides solutions for as many as four key design aspects. Other than the power management IC mentioned above, it provides UMTS transceiver, baseband processor and GPS receiver.

GPS. The in-built GPS capability is provided by Infineon's PMB2525 Hammerhead II GPS that is integrat-

Comparison between Different Versions of iPhone

	iPhone 2G	iPhone 3G	iPhone 3G S
CPU	Samsung S3C6400 ARM 11 412MHz, PowerVR MBX-Lite graphics	Samsung S3C6400 ARM 11 412MHz, PowerVR MBX-Lite graphics	Samsung S5PC100 ARM Cortex A8 600MHz PowerVR SGX graphics
Baseband processor	—	Infineon 208 (PMB 8877)	—
Storage	Samsung NAND internal 4/8/16 GB	Toshiba NAND internal 8/16 GB	Toshiba NAND internal 16/32 GB
Memory	128MB RAM	Samsung 128MB DDR RAM	256MB RAM
LCD	8.9cm (3.5-inch), 480×320 pixels at 163ppi multitouch display	8.9cm (3.5-inch), 480×320 pixels at 163ppi multitouch display	8.9cm (3.5-inch), 480×320 pixels at 163ppi multitouch display
Battery	Talk/video/web 8/7/6 hours Audio: 24 hours	Talk/video/web 5/7/5 hours Audio: 24 hours	Talk/video/web 5/10/9 hours Audio: 30 hours
Connectivity	GSM/EDGE (850, 900, 1800, 1900MHz) Wi-Fi (802.11b/g) Bluetooth 2.0+EDR	UMTS/HSDPA (850, 1900, 2100MHz) GSM/EDGE (850, 900, 1800, 1900MHz) Wi-Fi (802.11b/g) Bluetooth 2.0+EDR	UMTS/HSDPA (850, 1900, 2100MHz) GSM/EDGE (850, 900, 1800, 1900MHz) Wi-Fi (802.11b/g) Bluetooth 2.1+EDR
Size, weight	115×61×11.6 mm ³ 135 gm	115.5×62.1×12.3 mm ³ 133 gm	115.5×62.1×12.3 mm ³ 135 gm
Camera	2 megapixels	2 megapixels	3 megapixels

Note: Battery stats assume usage of 3G network, times vary with 2G network use for cellular and WiFi for Internet

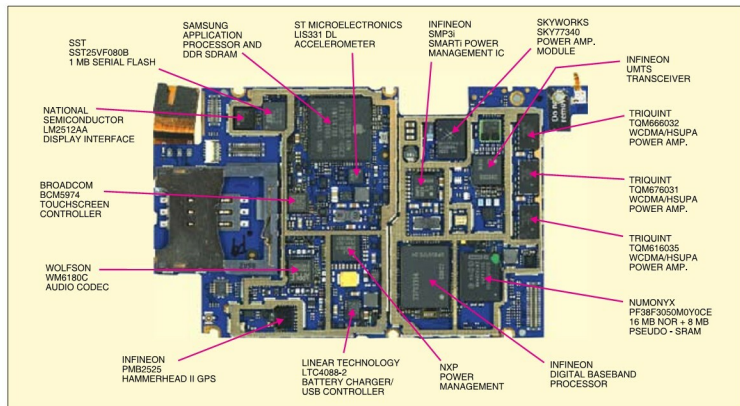


Fig. 2: Logic board of iPhone 3G showing the different ICs used